

Thermal Tolerance Hardware Evaluation

HARDWARE STRESS TEST AND THERMAL EVALUATION REPORT

Device Under Test: Silverblade X9 High-Density Server Chassis Test Date: August 15, 2026 Testing Engineer: Dr. Alan Vance

Objective: The purpose of this evaluation is to determine the thermal threshold and maximum sustained operating load of the Silverblade X9 chassis under extreme data center conditions. The test specifically measures CPU thermal throttling limits and cooling fan acoustic outputs at 100% simulated computational load.

Methodology: The chassis was equipped with dual 64-core processors and 512GB of DDR5 ECC RAM. We utilized the 'Prime95' benchmarking utility to generate maximum heat dissipation. The ambient room temperature of the testing chamber was strictly controlled at 25°C (77°F). Sensors were placed on the CPU heat sinks, VRM arrays, and exhaust vents.

Results: 1. Baseline Idle: At idle, CPU temperatures averaged 34°C. System fans operated at 1,200 RPM, producing a baseline acoustic profile of 38 dB(A).

2. Maximum Sustained Load (12 hours): Within 14 minutes of initiating Prime95, CPU package temperatures plateaued at 82°C. The internal logic successfully ramped the six hot-swappable cooling fans to their maximum threshold of 8,500 RPM. Acoustic output peaked at 74 dB(A), which is within the acceptable limits for server room deployment.

3. Thermal Throttling Event: To simulate a partial cooling failure, two of the six exhaust fans were intentionally disabled via software override at hour 13. Within 4 minutes, CPU temperatures breached the 95°C safety limit. The motherboard firmware correctly initiated thermal throttling, reducing the CPU clock speed from 4.2 GHz to 2.8 GHz to maintain a stable 92°C until the fans were re-enabled.

Conclusion: The Silverblade X9 chassis exhibits excellent thermal management under peak continuous loads. The firmware correctly intervenes during cooling degradation to prevent catastrophic hardware failure. The hardware is cleared for Tier 1 enterprise deployment.